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Housings for tracking devices

Abstract

Housings for tracking devices comprise a body and an electrical network. The body defines a device-receiving volume that is configured to selectively receive and retain the tracking device in the body, and a housing battery compartment that is configured to selectively receive and retain one or more housing batteries. The electrical network is operatively positioned within the body and is configured to electrically couple the one or more housing batteries with the tracking device.

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Background/Summary

RELATED APPLICATION (1) This application is a continuation of and claims priority to U.S. patent application Ser. No. 18/651,322, filed on Apr. 30, 2024, the disclosure of which is incorporated herein by reference.

FIELD

(1) The present disclosure relates to housings for tracking devices.

BACKGROUND

(2) In recent years, wireless tracking devices have become popular amongst consumers. Consumers often use tracking devices to keep track of sports equipment, electronic equipment, luggage, vehicles, etc. Typically, tracking devices are small and utilize batteries that last for only a limited period of time, such as six to twelve months or so. However, consumers may wish to track certain objects for longer periods of time without the need to replace the batteries in the tracking devices.

SUMMARY

(3) Housings for tracking devices comprise a body and an electrical network. The body defines a device-receiving volume that is configured to selectively receive and retain the tracking device in the body, and a housing battery compartment that is configured to selectively receive and retain one or more housing batteries. The electrical network is operatively positioned within the body and is configured to electrically couple the one or more housing batteries with the tracking device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic illustration representing housings for tracking devices according to the present disclosure.
- (2) FIG. 2 is a bottom isometric exploded view of an Apple™ AirTag™ device with a coin battery.
- (3) FIG. 3 is a top isometric view of an example housing according to the present disclosure.
- (4) FIG. 4 is a top isometric exploded view of the example housing of FIG. 3, shown with an Apple™ AirTag™ device and with housing batteries installed.
- (5) FIG. 5 is a further top isometric exploded view of the example housing of FIG. 3, shown with an Apple™ AirTag™ device and with housing batteries installed.
- (6) FIG. 6 is a top isometric exploded view of another example housing according to the present disclosure, shown with an Apple™ AirTag™ device and with housing batteries installed.

DESCRIPTION

(7) Housings **10** for tracking devices **12** and related methods are disclosed herein. Examples of devices **12** include wireless (e.g., WiFi, Bluetooth™, GPS) tracking devices (e.g., Apple™ AirTag™, Tile™, Samsung™ Galaxy™ SmartTag™, Chipolo™ One™, CubePro™, Orbit™, and Baseus™ branded tracking devices). Functionally, housings **10** may be used to power tracking devices **12** with one or more batteries that will last longer than the batteries typically (and/or intended to be) used with tracking devices **12**. For example, while a tracking device **12** may be designed for use with a small coin battery, such as a 3Volt (V) coin battery, a housing **10** may permit use of two AAA-, AA-, C-, or D-batteries rated at 1.5V in series with each other. Housings **10** may be configured for use with various suitable batteries depending on the particular tracking device **12** with which the housing **10** is intended to be used and/or depending on a desired length of time for the particular tracking device **12** to be powered. As an illustrative example, a 3V coin battery used with an Apple™ AirTag™ device may last about six months, while two 1.5V AA-batteries used with an Apple™ AirTag™ device in a housing **10** may last about ten years. Housings **10** additionally or alternatively may be described as or referred to as cases for tracking devices, battery packs for tracking devices, supplemental battery packs for tracking devices, battery banks for tracking devices, and the like.

(8) FIG. **1** schematically illustrates housings **10** according to the present disclosure. Generally in FIG. **1**, elements that are likely to be included in a given example are illustrated in solid lines, while elements that are optional to a given example or that correspond to a specific example are illustrated in broken lines. However, elements that are illustrated in solid lines are not essential to all examples of the present disclosure, and an element shown in solid lines may be omitted from a particular example without departing from the scope of the present disclosure.

(9) As schematically represented in FIG. **1**, housings **10** comprise at least a body **14** and an electrical network **22** that is operatively positioned within the body **14**. The body **14** defines a device-receiving volume **16** that is configured to selectively receive and retain a tracking device **12** in the body **14**. The body **14** further defines a housing battery compartment **18** configured to selectively receive and retain one or more housing batteries **20**. In some examples, the device-receiving volume **16** and the housing battery compartment **18** may be distinct from each other. In other examples, the two compartments generally may share the same volume within the body **14**. As noted above, depending on the particular tracking device **12** with which a housing **10** is to be used and depending on the desired life of the housing battery(ies) **20**, the housing battery(ies) **20** may comprise one or more various types and/or configurations of batteries. Such batteries may be non-rechargeable batteries (e.g., alkaline batteries) or rechargeable batteries (e.g., alkaline, Nickel-Cadmium [NiCd], Nickel-Metal Hydride [NiMH], or Lithium-ion [Li-ion] batteries). Various suitable sizes of housing batteries **20** include (but are not limited to) AAA-, AA-, C-, and D-batteries.

(10) As schematically represented in FIG. **1**, the electrical network **22** is configured to electrically couple the housing battery(ies) **20** with the tracking device **12** when the tracking device **12** is operatively received in the device-receiving volume **16** and when the housing battery(ies) **20** are operatively received in the housing battery compartment **18**. The electrical network **22** in combination with the housing battery(ies) **20** and the tracking device **12** complete an electrical circuit for operatively powering the tracking device **12**.

(11) With continued reference to the schematic representation of FIG. **1** and also with reference to the example tracking device **12** of FIG. **2** in the form of an Apple™ AirTag™ device **12'**, in some examples, the tracking device **12** defines a tracking-device battery compartment **24** that is configured to operatively receive one or more device batteries **48**, such as a coin battery **48'**, as in the example of FIG. **2**. In such examples, the tracking device **12** comprises a first tracking-device contact **26** that is positioned to operatively engage a first terminal **50** of the device battery(ies) **48** when the device battery(ies) **48** are operatively received in the tracking-device battery compartment

24. Similarly in such examples, the tracking device **12** comprises a second tracking-device contact **28** that is positioned to operatively engage a second terminal **52** of the device battery(ies) **48** when the device battery(ies) **48** are operatively received in the tracking-device battery compartment **24**. In the example of an Apple™ AirTag™ device **12'**, two first tracking-device contacts **26'** provide operative contact with the first terminal **50** of a coin battery **48'**. The first tracking-device contact(s) **26** and the second tracking-device contact(s) **28** collectively may be referred to herein as the tracking-device's contacts.

(12) Herein, "Apple™ AirTag™ device" refers to the tracking device illustrated in FIG. 2 and available in the marketplace from Apple Inc. under the AirTag™ trademark as of the date of the present application. "Apple™ AirTag™ device" also refers to future variations thereon that have a similar structure as illustrated in FIG. 2, such that housings **10** as disclosed herein could operatively receive such a tracking device **12** for powering of the tracking device **12**.

(13) As schematically represented in FIG. 1, in some examples, the electrical network **22** of the housing **10** comprises a first electrical contact **30** and a second electrical contact **32**. The first electrical contact **30** is positioned to operatively engage the first tracking-device contact **26** when the tracking device **12** is operatively received in the device-receiving volume **16**, and the second electrical contact **32** is positioned to operatively engage the second tracking-device contact **28** when the tracking device **12** is operatively received in the device-receiving volume **16**. The first electrical contact(s) **30** and the second electrical contact(s) **32** collectively may be referred to herein as the housing's contacts.

(14) In some examples of tracking device **12**, such as an Apple™ AirTag™ device **12'**, the first tracking-device contact(s) **26** and the second tracking-device contact **28** are spring-biased toward, or into, the tracking-device battery compartment **24**. Such contacts additionally or alternatively may be referred to as, or as comprising, pogos or pogo pins. In such examples, when a device battery **48** is operatively positioned within the tracking-device battery compartment **24**, and a battery cover **46** of the tracking device **12** (see example tracking device **12'** of FIG. 2) is operatively installed, the device battery **48** engages and urges the pogo pins against their spring biases.

(15) With continued reference to FIG. 1, some tracking devices **12** may be described as having a central axis **42**. In some examples of housings **10**, the housing **10** is configured to operatively receive the tracking device **12** in any radial orientation of the tracking device **12** about the central axis **42**, and to have the first electrical contact **30** operatively engage the first tracking-device contact **26** and the second electrical contact **32** operatively engage the second tracking-device contact **28** when the tracking device **12** is operatively received in the device-receiving volume **16**. Accordingly, in such examples of housings **10**, a user need not worry about aligning the tracking device **12** in a particular radial orientation about the central axis **42**, as any radial orientation will result in operative engagement of the tracking device's contacts with the housing's contacts.

(16) In some examples, the second electrical contact **32** comprises a disc-shaped body **36** (i.e., a generally cylindrical shape with its radius greater than its thickness). As a result, the disc-shaped body **36** can make operative contact with a corresponding second tracking-device contact **28** regardless of the radial orientation of the tracking device **12**, and thus regardless of where the second tracking-device contact **28** is positioned within the tracking-device battery compartment **24** for operative contact at any position on the disc-shaped body **36**.

(17) In some examples, the first electrical contact **30** comprises a ring-shaped body **34**, that is a generally circular body that extends greater than 180-degrees, greater than 270-degrees, greater than 305-degrees, or even a full 360-degrees about a central axis. As a result, the ring-shaped body **34** can make operative contact with a corresponding first tracking-device contact **26** regardless of the radial orientation of the tracking device **12**, and thus regardless of where the first tracking-device contact **26** is positioned within the tracking-device battery compartment **24** for operative contact at any position on the ring-shaped body **34**.

(18) In some such examples, the ring-shaped body **34** and the disc-shaped body **36** are coaxial with each other. In particular, the disc-shaped body **36** and the ring-shaped body **34** may generally correspond to the shape and/or size of the terminals of a coin battery configured for use with a particular tracking device **12** (e.g., coin battery **48'** of FIG. 2). In some such examples, the ring-shaped body **34** and the disc-shaped body **36** also are coaxial with the central axis **42** of the tracking device **12** when the tracking device **12** is operatively received in the device-receiving volume **16**.

(19) With continued reference to the schematic representation of FIG. 1, but also with reference to the specific example of an Apple™ AirTag™ device **12'** of FIG. 2, some tracking devices **12** comprise one or more structural features **38** that at least partially define the tracking-device battery compartment **24** or that extend into the tracking-device battery compartment **24**. When a housing **10** is configured for use with such a tracking device **12**, some such housings **10** may comprise one or more indexing features **40** that are configured to engage the structural feature(s) **38** of the tracking device **12** for alignment and engagement of the first electrical contact **30** with the first tracking-device contact **26**, and for alignment and engagement of the second electrical contact **32** with the second tracking-device contact **28** when the tracking device **12** is operatively received in the device-receiving volume **16**. In other words, in contrast to the examples discussed above, in which the housing **10** is configured to operatively receive the tracking device **12** in any radial orientation of the tracking device **12** about the central axis **42**, other housings **10** may be configured to operatively mate with a tracking device **12** in only one or more specific orientations dictated by the indexing features **40** to ensure that the tracking device's contacts operatively engage the housing's contacts.

(20) In some such examples, such as the example Apple™ AirTag™ device **12'** of FIG. 2, the structural feature(s) **38** are configured to selectively mate with the battery cover **46** of the tracking device **12** that is configured to selectively retain the device battery(ies) **48** within the tracking-device battery compartment **24**. That is, the one or more structural features **38** may serve as mating features for the associated battery cover **46**, such as in the form of threads, fastener holes, or other fastener structures, friction-fit structures, snap-fit structures, and the like.

(21) In further examples, the device-receiving volume **16** of a housing **10** may be configured to operatively mate with the structural feature(s) **38** of a tracking device **12** to ensure operative alignment and engagement between the housing's contacts and the tracking device's contacts. For example, the body **14** of a housing **10** may comprise threads or other structure(s) (e.g., friction-fit structure(s), snap-fit structure(s), etc.) that are configured to mate with corresponding structural feature(s) **38** in the form of threads or other structures (e.g., friction-fit structure(s), snap-fit structure(s), etc.).

(22) With reference back to FIG. 1 and as represented by the dashed lines generally bisecting the solid rectangular schematic representation of the body **14**, in some examples of housings **10**, the body **14** comprises a base **56** and one or more covers **58** that are removably coupled to the base **56** to permit placement and removal of the tracking device **12** in and from the device-receiving volume **16**, and to permit placement and removal of the housing battery(ies) **20** in and from the housing battery compartment **18**. For example, a single cover **58** may provide access to both the device-receiving volume **16** and the housing battery compartment **18**. In other examples, separate covers **58** may provide respective access to the device-receiving volume **16** and the housing battery compartment **18**. The cover(s) **58** may be removably coupled to the base **56** via any suitable mechanism, such as but not limited to, fasteners (e.g., screws), threads, friction-fit structures, snap-fit structures, etc.

(23) In some examples and as schematically represented in FIG. 1, housings **10** further comprise one or more gaskets **60** that are operatively positioned to provide a seal or seals between the base **56** and the cover(s) **58** when the cover(s) **58** are operatively coupled to the base **56**. Such housings **10** may be described as or configured to be waterproof for operative protection of the housing

battery(ies) **20**, the tracking device **12**, and the electrical network **22**.

(24) In some examples, the base **56** comprises a main base **62** and a base insert **64** that is operatively coupled to the main base **62** and that at least partially defines the device-receiving volume **16**. In some such examples, the electrical network **22** comprises wires **66**, and the base insert **64** at least partially conceals the wires **66** between the base insert **64** and the main base **62**. In some examples, the base insert **64** is electrically non-conductive. In some examples, at least a portion of the electrical network **22**, such as one or both of the first electrical contact **30** and/or the second electrical contact **32**, extends through the base insert **64**. The base insert **64** may be operatively coupled to the main base **62** in any suitable manner, such as with fasteners or adhesive.

(25) In some examples of housings **10** configured for use with tracking devices **12** that define a tracking-device battery compartment **24**, the base **56** of the housing **10** may comprise a projection **68** that is positioned and configured to operatively extend into the tracking-device battery compartment **24** when the tracking device **12** is operatively received in the device-receiving volume **16**. That is, the base **56** may provide structure for operative mating of the base **56** with a tracking device **12**, including for the operative engagement of the housing's contacts with the tracking device's contacts. In some such examples, the base insert **64** comprises the projection **68**. Accordingly, in such examples, the base insert **64** not only may conceal portions of the electrical network, but it also may provide structure for operative mating of the housing **10** with a tracking device **12** and operative engagement of the housing's contacts with the tracking device's contacts.

(26) In some examples, the projection **68** is generally cylindrical, and thus is configured for mating with a generally cylindrical or otherwise circular tracking-device battery compartment **24**, such as in the example of the Apple™ AirTag™ device **12'** of FIG. 2.

(27) In some examples, the projection **68** at least partially supports the first electrical contact **30** and the second electrical contact **32**. That is, the projection **68** may provide support for operative and continued engagement between the housing's contacts and the tracking device's contacts when the tracking device **12** is operatively mated with the projection **68**. In some examples, the projection **68** is electrically non-conductive to avoid the housing's contacts shorting with each other.

(28) As schematically represented in FIG. 1, in some examples, the projection **68** comprises a perimeter wall **70** and an end wall **72**. In some such examples, the first electrical contact **30** is at least partially supported by the perimeter wall **70**, and the second electrical contact **32** is at least partially supported by the end wall **72**. For example, in examples where the first electrical contact **30** comprises a ring-shaped body **34**, the perimeter wall **70** of the projection **68** may support the ring-shaped body **34**; and in examples where the second electrical contact **32** comprises a disc-shaped body **36**, the end wall **72** of the projection **68** may support the disc-shaped body **36**. In some such examples, the end wall **72** defines a recess **74**, and the second electrical contact **32** (e.g., the disc-shaped body **36** thereof) is positioned at least partially within the recess **74**. Accordingly, the projection **68** may provide insulation around an operative portion of the second electrical contact **32** to restrict shorting of the second electrical contact **32** with the first electrical contact **30**. In examples where the second electrical contact **32** comprises a disc-shaped body **36**, the recess may be circular for operative receipt of the disc-shaped body **36**.

(29) In some examples, at least a portion of the electrical network **22** extends through the end wall **72** of the projection. For example, leads of the second electrical contact **32**, or leads or wires operatively connecting the second electrical contact **32** to a contact associated with the housing battery(ies) **20**, may extend through the end wall **72** of the projection.

(30) In some examples, rather than being supported by the projection **68**, the housing's contacts may be described as at least partially defining the projection **68**.

(31) In some examples, the device-receiving volume **16** is configured to compress the first tracking-device contact **26** and/or the second tracking-device contact **28** when the tracking device **12** is operatively received in the device-receiving volume **16**. Such a configuration may ensure

operative and continuous contact between the housing's contacts and the tracking device's contacts, preventing the tracking device from moving within the device-receiving volume **16**. In some such examples, when the tracking-device's contacts are spring-biased, the device-receiving volume **16** is configured to compress the tracking device's contacts when the tracking device **12** is operatively received in the device-receiving volume **16**.

(32) In some examples as schematically represented in FIG. **1**, the body **14** comprises one or more ribs **76** that are operatively positioned to engage the tracking device **12** when the tracking device **12** is operatively received in the device-receiving volume **16**. In some such examples, a cover **58** associated with the device-receiving volume **16** comprises the rib(s) **76**. In some examples, when the tracking device's contacts are spring-biased, the rib(s) **76** are configured to operatively compress the first tracking-device contact **26** and/or the second tracking-device contact **28** (i.e., cause the first tracking-device contact **26** and/or the second tracking-device contact **28** to become compressed) when the rib(s) **76** operatively engage the tracking device **12**, when the tracking device **12** is operatively received in the device-receiving volume **16**. That is, the rib(s) **76** may push against the tracking device **12** to ensure a snug fit of the tracking device **12** within the device-receiving volume **16** and to ensure operative contact between the housing's contacts and the tracking device's contacts. In some examples, the rib(s) **76** are positioned to engage only a peripheral region of a surface of the tracking device **12** when the tracking device **12** is operatively received within the device-receiving volume **16**. Such a configuration of rib(s) **76** may avoid muffling an internal speaker of the tracking device **12**. That is, by only engaging a peripheral region of the tracking device **12**, a volume of air is provided directly adjacent to the tracking device **12**, permitting sound waves to transfer to the body **14** of the housing **10** and ultimately into the surrounding environment.

(33) In some examples, the rib(s) **76** comprise a circular rib **78** that is positioned to engage only a peripheral region of tracking device **12**. In some such examples, the circular rib **78**, when present, is coaxial with the ring-shaped body **34** and the disc-shaped body **36** of the housing's contacts, when so configured. Additionally or alternatively, in examples that comprise a projection **68**, the circular rib **78** may be coaxial with the projection **68**.

(34) As schematically represented in FIG. **1**, some housings **10** further comprise a coupler **80** that is configured to operatively attach the housing **10** to an article. In FIG. **1**, the optional coupler **80** is schematically represented in an overlapping relationship with the body **14**, schematically representing that the coupler **80** may be coupled to an outer surface of the body **14**, may be positioned inside of the body **14**, or may be integral with the body **14**. The coupler **80** may take various configurations, including for example, an adhesive, a double-sided tape, a double-sided foam tape, one or more magnets, fasteners, fastener structures (e.g., holes), etc.

(35) Also within the scope of the present disclosure are methods of powering a tracking device **12**. Such methods comprise: powering the tracking device with a battery cover **46** of the tracking device **12** removed and with no device batteries **48** within a tracking-device battery compartment **24** of the tracking device **12**; operatively engaging a first electrical contact **30** of a housing **10** with a first tracking-device contact **26** within the tracking-device battery compartment **24** of the tracking device **12**; and operatively engaging a second electrical contact **32** of the housing **10** with a second tracking-device contact **28** within the tracking-device battery compartment **24** of the tracking device **12**.

(36) Turning now to FIGS. **3-6**, non-exclusive examples of housings **10** in the form of housing **100** and housing **200** are illustrated. Where appropriate, the reference numerals from the schematic illustration of FIG. **1** are used with one or more prime symbols (') to designate corresponding parts of housing **100** and housing **200**; however, the examples of FIGS. **3-6** are non-exclusive and do not limit housings **10** to the illustrated embodiments of housing **100** and housing **200**. That is, housings **10** are not limited to the specific embodiments of the illustrated housing **100** and housing **200**, and housings **10** may incorporate any number of the various aspects, configurations, characteristics,

properties, etc. of housings **10** that are illustrated and discussed with reference to the schematic representation of FIG. **1** and/or the embodiments of FIGS. **3-6**, as well as variations thereof, without requiring the inclusion of all such aspects, configurations, characteristics, properties, etc. For the purpose of brevity, each previously discussed component, part, portion, aspect, region, etc., or variants thereof may not be discussed, illustrated, and/or labeled again with respect to housing **100** and housing **200**. However, it is within the scope of the present disclosure that the previously discussed features, variants, etc. may be utilized with housing **100** or housing **200**.

(37) FIGS. **3-5** illustrate example housing **100**. Housing **100** is an example of a housing **10** that is configured to power an Apple™ AirTag™ device **12'** utilizing two AA-batteries. In particular, housing **100** is configured to receive an Apple™ AirTag™ device **12'** with its battery cover **46** removed, and with the Apple™ AirTag™ device **12'** in any radial orientation about its central axis. The body **14'** of housing **100** comprises a base **56'** and a cover **58'**. The base **56'** comprises a main base **62'** and a base insert **64'**. The base insert **64'** comprises a generally cylindrical projection **68'** that supports the housing's contacts. As perhaps best seen in the exploded view of FIG. **5**, the projection **68'** of housing **100** comprises a perimeter wall **70'** that supports the first electrical contact **30'**, and an end wall **72'** that supports the second electrical contact **32'**. Moreover, housing **100** is an example of a housing **10** whose first electrical contact **30'** comprises a ring-shaped body **34'** and whose second electrical contact **32'** comprises a disc-shaped body **36'**. The end wall **72'** defines a recess **74'**, within which the disc-shaped body **36'** is seated. The first electrical contact **30'** also comprises leads **102** that extend from the ring-shaped body **34'**, through the base insert **64'**. At least one of the leads **102** is coupled to a wire **66** (not depicted) for operative connection to the batteries **20'**. The second electrical contact **32'** also comprises leads **104** that extend from the disc-shaped body **36'** and through the end wall **72'** of the projection **68'** of the base insert **64'**. At least one of the leads **104** is coupled to a wire **66** (not depicted) for operative connection to the batteries **20'**.

(38) FIG. **6** illustrates example housing **200**. Housing **200** is another example of a housing **10** that is configured to power an Apple™ AirTag™ device **12'** utilizing two AA-batteries. Like housing **100**, the body **14''** of housing **200** comprises a base **56''** and a cover **58''**, and the base **56''** comprises a main base **62''** and a base insert **64''**. Also like housing **100**, housing **200** is configured to receive an Apple™ AirTag™ device **12'** with its battery cover **46** removed. However, unlike housing **100**, housing **200** is not configured to receive an Apple™ AirTag™ device **12'** in any radial orientation about its central axis. Rather, housing **200** is an example of a housing **10** that is configured to receive an Apple™ AirTag™ device **12'** in a single radial orientation, to ensure that the housing's contacts operatively engage the tracking device's contacts. In particular, as seen in FIG. **6**, the base insert **64''** of housing **200** comprises three ribs **202** that serve as indexing features **40''** for engagement with corresponding structural features within the tracking-device battery compartment **24'** of the Apple™ AirTag™ device **12'**.

(39) Illustrative, non-exclusive examples of inventive subject matter according to the present disclosure are described in the following enumerated paragraphs:

(40) A. A housing (**10**) for a tracking device (**12**), the housing (**10**) comprising: a body (**14**), wherein the body (**14**) defines: a device-receiving volume (**16**) configured to selectively receive and retain the tracking device (**12**) in the body (**14**); and a housing battery compartment (**18**) configured to selectively receive and retain one or more housing batteries (**20**); and an electrical network (**22**) operatively positioned within the body (**14**) and configured to electrically couple the one or more housing batteries (**20**) with the tracking device (**12**) when the tracking device (**12**) is operatively received in the device-receiving volume (**16**) and when the one or more housing batteries (**20**) are operatively received in the housing battery compartment (**18**).

(41) A1. The housing (**10**) of paragraph A, wherein the tracking device (**12**) defines a tracking-device battery compartment (**24**) configured to operatively receive one or more device batteries (**48**); wherein the tracking device (**12**) comprises: a first tracking-device contact (**26**) positioned to

operatively engage a first terminal (50) of the one or more device batteries (48) when the one or more device batteries (48) are operatively received in the tracking-device battery compartment (24); and a second tracking-device contact (28) positioned to operatively engage a second terminal (52) of the one or more device batteries (48) when the one or more device batteries (48) are operatively received in the tracking-device battery compartment (24); and wherein the electrical network (22) comprises: a first electrical contact (30) positioned to operatively engage the first tracking-device contact (26) when the tracking device (12) is operatively received in the device-receiving volume (16); and a second electrical contact (32) positioned to operatively engage the second tracking-device contact (28) when the tracking device (12) is operatively received in the device-receiving volume (16).

(42) A1.1. The housing (10) of paragraph A1, wherein the one or more device batteries (48) comprise one or more coin batteries (54).

(43) A1.2. The housing (10) of any of paragraphs A1-A1.1, wherein the first tracking-device contact (26) and the second tracking-device contact (28) are spring-biased toward the tracking-device battery compartment (24).

(44) A1.3. The housing (10) of any of paragraphs A1-A1.2, wherein the tracking device (12) has a central axis (42); and wherein the housing (10) is configured to operatively receive the tracking device (12) in any radial orientation of the tracking device (12) about the central axis (42), and have the first electrical contact (30) operatively engage the first tracking-device contact (26) and the second electrical contact (32) operatively engage the second tracking-device contact (28) when the tracking device (12) is operatively received in the device-receiving volume (16).

(45) A1.4. The housing (10) of any of paragraphs A1-A1.3, wherein the second electrical contact (32) comprises a disc-shaped body (36).

(46) A1.5. The housing (10) of any of paragraphs A1-A1.4, wherein the first electrical contact (30) comprises a ring-shaped body (34).

(47) A1.5.1. The housing (10) of paragraph A1.5 when depending from paragraph A1.4, wherein the ring-shaped body (34) and the disc-shaped body (36) are coaxial.

(48) A1.5.1.1. The housing (10) of paragraph A1.5.1 when depending from paragraph A1.3, wherein the ring-shaped body (34), the disc-shaped body (36), and the central axis (42) of the tracking device (12) are coaxial when the tracking device (12) is operatively received in the device-receiving volume (16).

(49) A1.6. The housing (10) of any of paragraphs A1-A1.2, wherein the tracking device (12) comprises one or more structural features (38) that at least partially define the tracking-device battery compartment (24) or that extend into the tracking-device battery compartment (24); and wherein the housing (10) further comprises one or more indexing features (40) configured to engage the one or more structural features (38) for alignment and engagement of the first electrical contact (30) with the first tracking-device contact (26) and for alignment and engagement of the second electrical contact (32) with the second tracking-device contact (28) when the tracking device (12) is operatively received in the device-receiving volume (16).

(50) A1.6.1. The housing (10) of paragraph A1.6, wherein the one or more structural features (38) are configured to selectively mate with a battery cover (46) that is configured to selectively retain the one or more device batteries (48) within the tracking-device battery compartment (24).

(51) A2. The housing (10) of any of paragraphs A-A1.6.1, wherein the body (14) comprises: a base (56); and one or more covers (58) removably coupled to the base (56) to permit placement and removal of the tracking device (12) in and from the device-receiving volume (16) and to permit placement and removal of the one or more housing batteries (20) in and from the housing battery compartment (18).

(52) A2.1. The housing (10) of paragraph A2, further comprising: one or more gaskets (60) operatively positioned to provide a seal or seals between the base (56) and the one or more covers (58) when the one or more covers (58) are operatively coupled to the base (56).

(53) A2.2. The housing (10) of any of paragraphs A2-A2.1, wherein the base (56) comprises: a main base (62); and a base insert (64) operatively coupled to the main base (62) and at least partially defining the device-receiving volume (16).

(54) A2.2.1. The housing (10) of paragraph A2.2, wherein the electrical network (22) comprises wires (66); and wherein the base insert (64) at least partially conceals the wires (66) between the base insert (64) and the main base (62).

(55) A2.2.2. The housing (10) of any of paragraphs A2.2-A2.2.1, wherein the base insert (64) is electrically non-conductive.

(56) A2.2.3. The housing (10) of any of paragraphs A2.2-A2.2.2, wherein at least a portion of the electrical network (22) extends through the base insert (64).

(57) A2.3. The housing (10) of any of paragraphs A2-A2.2.3, wherein the tracking device (12) defines a/the tracking-device battery compartment (24) configured to operatively receive (the) one or more device batteries (48); and wherein the base (56) comprises a projection (68) positioned to operatively extend into the tracking-device battery compartment (24) when the tracking device (12) is operatively received in the device-receiving volume (16).

(58) A2.3.1. The housing (10) of paragraph A2.3 when depending from paragraph A2.2, wherein the base insert (64) comprises the projection (68).

(59) A2.3.2. The housing (10) of any of paragraphs A2.3-A2.3.1, wherein the projection (68) is generally cylindrical.

(60) A2.3.3. The housing (10) of any of paragraphs A2.3-A2.3.2 when depending from paragraph A1, wherein the projection (68) at least partially supports the first electrical contact (30) and the second electrical contact (32).

(61) A2.3.4. The housing (10) of any of paragraphs A2.3-A2.3.3, wherein the projection (68) is electrically non-conductive.

(62) A2.3.5. The housing (10) of any of paragraphs A2.3-A2.3.4, wherein the projection (68) comprises a perimeter wall (70) and an end wall (72), wherein the first electrical contact (30) is at least partially supported by the perimeter wall (70), and wherein the second electrical contact (32) is at least partially supported by the end wall (72).

(63) A2.3.5.1. The housing (10) of paragraph A2.3.5, wherein the end wall (72) defines a recess (74), and wherein the second electrical contact (32) is positioned at least partially within the recess (74).

(64) A2.3.5.1.1. The housing (10) of paragraph A2.3.5.1 when depending from paragraph A2.3.2, wherein the recess (74) is circular.

(65) A2.3.5.2. The housing (10) of any of paragraphs A2.3.5-A2.3.5.1.1, wherein at least a portion of the electrical network (22) extends through the end wall (72).

(66) A3. The housing (10) of any of paragraphs A-A2.3.5.2, wherein the tracking device (12) comprises: a/the first tracking-device contact (26) positioned to operatively engage a/the first terminal (50) of (the) one or more device batteries (48) when the one or more device batteries (48) are operatively received in a/the tracking-device battery compartment (24), wherein the first tracking-device contact (26) is spring-biased; and a/the second tracking-device contact (28) positioned to operatively engage a/the second terminal (52) of the one or more device batteries (48) when the one or more device batteries (48) are operatively received in the tracking-device battery compartment (24), wherein the second tracking-device contact (28) is spring-biased; and wherein the device-receiving volume (16) is configured to compress the first tracking-device contact (26) and the second tracking-device contact (28) when the tracking device (12) is operatively received in the device-receiving volume (16).

(67) A4. The housing (10) of any of paragraphs A-A3, wherein the body (14) comprises one or more ribs (76) operatively positioned to engage the tracking device (12) when the tracking device (12) is operatively received in the device-receiving volume (16).

(68) A4.1. The housing (10) of paragraph A4 when depending from paragraph A2, wherein the one

or more covers (58) comprise the one or more ribs (76).

(69) A4.2. The housing (10) of any of paragraphs A4-A4.1 when depending from paragraph A1.2, wherein the one or more ribs (76) are configured to operatively compress the first tracking-device contact (26) and/or the second tracking-device contact (28) when the one or more ribs (76) operatively engage the tracking device (12) when the tracking device (12) is operatively received in the device-receiving volume (16).

(70) A4.3. The housing (10) of any of paragraphs A4-A4.2, wherein the one or more ribs (76) comprise a circular rib (78).

(71) A4.3.1. The housing (10) of paragraph A4.3 when depending from paragraph A1.5.1, wherein the circular rib (78) is coaxial with the ring-shaped body (34) and the disc-shaped body (36).

(72) A4.3.2. The housing (10) of any of paragraphs A4.3-A4.3.1 when depending from paragraph A2.3, wherein the circular rib (78) is coaxial with the projection (68).

(73) A5. The housing (10) of any of paragraphs A-A4.3.2, further comprising: a coupler (80) configured to operatively attach the housing (10) to an article (82).

(74) A6. The housing (10) of any of paragraphs A-A5, wherein the device-receiving volume (16) is configured to selectively receive and retain the tracking device (12) in the body (14) without a/the battery cover (46) of the tracking device (12) present.

(75) A7. The housing (10) of any of paragraphs A-A6, further comprising the tracking device (12).

(76) A7.1. The housing (10) of paragraph A6, wherein the tracking device (12) is operatively received in the device-receiving volume (16).

(77) A8. The housing (10) of any of paragraphs A-A7.1, wherein the tracking device (12) comprises an Apple™ AirTag™ device (12').

(78) B. A housing (10) for a tracking device (12), the housing (10) comprising: a body (14), wherein the body (14) defines: a device-receiving volume (16) configured to selectively receive and retain the tracking device (12) in the body (14); and a housing battery compartment (18) configured to selectively receive and retain one or more housing batteries (20); and means for electrically connecting the tracking device (12) to the one or more housing batteries (20).

(79) B1. The housing (10) of paragraph B, wherein the means for electrically connecting comprises the subject matter of any of paragraphs A-A8.

(80) C. A method of powering a tracking device (12), the method comprising: with a battery cover (46) of the tracking device (12) removed and with no device batteries (48) within a tracking-device battery compartment (24) of the tracking device (12): operatively engaging a first electrical contact (30) of a housing (10) with a first tracking-device contact (26) within the tracking-device battery compartment (24) of the tracking device (12); and operatively engaging a second electrical contact (32) of the housing (10) with a second tracking-device contact (28) within the tracking-device battery compartment (24) of the tracking device (12).

(81) C1. The method of paragraph C, wherein the housing (10) comprises the housing (10) of any of paragraphs A-B1.

(82) D. Use of the housing (10) of any of paragraphs A-B1 to power the tracking device (12).

(83) As used herein, the terms “adapted” and “configured” mean that the element, component, or other subject matter is designed and/or intended to perform a given function. Thus, the use of the terms “adapted” and “configured” should not be construed to mean that a given element, component, or other subject matter is simply “capable of” performing a given function but that the element, component, and/or other subject matter is specifically selected, created, implemented, utilized, programmed, and/or designed for the purpose of performing the function. It is also within the scope of the present disclosure that elements, components, and/or other recited subject matter that is recited as being adapted to perform a particular function may additionally or alternatively be described as being configured to perform that function, and vice versa. Similarly, subject matter that is recited as being configured to perform a particular function may additionally or alternatively be described as being operative to perform that function.

(84) As used herein, the term “and/or” placed between a first entity and a second entity means one of (1) the first entity, (2) the second entity, and (3) the first entity and the second entity. Multiple entries listed with “and/or” should be construed in the same manner, i.e., “one or more” of the entities so conjoined. Other entities optionally may be present other than the entities specifically identified by the “and/or” clause, whether related or unrelated to those entities specifically identified. Thus, as a non-limiting example, a reference to “A and/or B,” when used in conjunction with open-ended language such as “comprising,” may refer, in one example, to A only (optionally including entities other than B); in another example, to B only (optionally including entities other than A); in yet another example, to both A and B (optionally including other entities). These entities may refer to elements, actions, structures, steps, operations, values, and the like.

(85) The various disclosed elements of apparatuses and steps of methods disclosed herein are not required to all apparatuses and methods according to the present disclosure, and the present disclosure includes all novel and non-obvious combinations and subcombinations of the various elements and steps disclosed herein. Moreover, one or more of the various elements and steps disclosed herein may define independent inventive subject matter that is separate and apart from the whole of a disclosed apparatus or method. Accordingly, such inventive subject matter is not required to be associated with the specific apparatuses and methods that are expressly disclosed herein, and such inventive subject matter may find utility in apparatuses and/or methods that are not expressly disclosed herein.

Claims

1. A housing (10) for a tracking device (12) that defines a tracking-device battery compartment (24) configured to operatively receive one or more device batteries (48), wherein the tracking device (12) comprises a first tracking-device contact (26) positioned to operatively engage a first terminal (50) of the one or more device batteries (48) when the one or more device batteries (48) are operatively received in the tracking-device battery compartment (24), and a second tracking-device contact (28) positioned to operatively engage a second terminal (52) of the one or more device batteries (48) when the one or more device batteries (48) are operatively received in the tracking-device battery compartment (24), wherein the housing (10) comprises: a body (14), wherein the body (14) defines: a device-receiving volume (16) configured to selectively receive and retain the tracking device (12) in the body (14) without a battery cover (46) of the tracking device (12) present; and a housing battery compartment (18) configured to selectively receive and retain one or more housing batteries (20); and an electrical network (22) operatively positioned within the body (14) and configured to electrically couple the one or more housing batteries (20) with the tracking device (12) when the tracking device (12) is operatively received in the device-receiving volume (16) and when the one or more housing batteries (20) are operatively received in the housing battery compartment (18), wherein the electrical network (22) comprises: a first electrical contact (30) positioned to operatively engage the first tracking-device contact (26) when the tracking device (12) is operatively received in the device-receiving volume (16); and a second electrical contact (32) positioned to operatively engage the second tracking-device contact (28) when the tracking device (12) is operatively received in the device-receiving volume (16).

2. The housing (10) of claim 1, wherein the tracking device (12) has a central axis (42); and wherein the housing (10) is configured to operatively receive the tracking device (12) in any radial orientation of the tracking device (12) about the central axis (42), and have the first electrical contact (30) operatively engage the first tracking-device contact (26) and the second electrical contact (32) operatively engage the second tracking-device contact (28) when the tracking device (12) is operatively received in the device-receiving volume (16).

3. The housing (10) of claim 1, wherein the second electrical contact (32) comprises a disc-shaped body (36).

4. The housing (10) of claim 3, wherein the first electrical contact (30) comprises a ring-shaped body (34).
5. The housing (10) of claim 4, wherein the ring-shaped body (34) and the disc-shaped body (36) are coaxial.
6. The housing (10) of claim 4, wherein the tracking device (12) has a central axis (42); wherein the housing (10) is configured to operatively receive the tracking device (12) in any radial orientation of the tracking device (12) about the central axis (42), and have the first electrical contact (30) operatively engage the first tracking-device contact (26) and the second electrical contact (32) operatively engage the second tracking-device contact (28) when the tracking device (12) is operatively received in the device-receiving volume (16); and wherein the ring-shaped body (34), the disc-shaped body (36), and the central axis (42) of the tracking device (12) are coaxial when the tracking device (12) is operatively received in the device-receiving volume (16).
7. The housing (10) of claim 1, wherein the first electrical contact (30) comprises a ring-shaped body (34).
8. The housing (10) of claim 1, wherein the tracking device (12) comprises one or more structural features (38) that are configured to selectively mate with the battery cover (46), wherein the battery cover (46) is configured to selectively retain the one or more device batteries (48) within the tracking-device battery compartment (24); and wherein the housing (10) further comprises one or more indexing features (40) configured to engage the one or more structural features (38) for alignment and engagement of the first electrical contact (30) with the first tracking-device contact (26) and for alignment and engagement of the second electrical contact (32) with the second tracking-device contact (28) when the tracking device (12) is operatively received in the device-receiving volume (16).
9. The housing (10) of claim 1, wherein the body (14) comprises: a base (56); and one or more covers (58) removably coupled to the base (56) to permit placement and removal of the tracking device (12) in and from the device-receiving volume (16) and to permit placement and removal of the one or more housing batteries (20) in and from the housing battery compartment (18).
10. The housing (10) of claim 9, further comprising: one or more gaskets (60) operatively positioned to provide a seal or seals between the base (56) and the one or more covers (58) when the one or more covers (58) are operatively coupled to the base (56).
11. The housing (10) of claim 9, wherein the base (56) comprises: a main base (62); and a base insert (64) operatively coupled to the main base (62) and at least partially defining the device-receiving volume (16).
12. The housing (10) of claim 11, wherein the electrical network (22) comprises wires (66); and wherein the base insert (64) at least partially conceals the wires (66) between the base insert (64) and the main base (62).
13. The housing (10) of claim 11, wherein the base insert (64) is electrically non-conductive, and wherein at least a portion of the electrical network (22) extends through the base insert (64).
14. The housing (10) of claim 11, wherein the base (56) comprises a projection (68) positioned to operatively extend into the tracking-device battery compartment (24) when the tracking device (12) is operatively received in the device-receiving volume (16).
15. The housing (10) of claim 14, wherein the base insert (64) comprises the projection (68).
16. The housing (10) of claim 14, wherein the projection (68) is generally cylindrical.
17. The housing (10) of claim 14, wherein the projection (68) at least partially supports the first electrical contact (30) and the second electrical contact (32).
18. The housing (10) of claim 17, wherein the projection (68) comprises a perimeter wall (70) and an end wall (72), wherein the first electrical contact (30) is at least partially supported by the perimeter wall (70), and wherein the second electrical contact (32) is at least partially supported by the end wall (72).
19. The housing (10) of claim 18, wherein the end wall (72) defines a recess (74), and wherein the

second electrical contact (32) is positioned at least partially within the recess (74).

20. The housing (10) of claim 19, wherein at least a portion of the electrical network (22) extends through the end wall (72).

21. The housing (10) of claim 1, wherein the device-receiving volume (16) is configured to compress the first tracking-device contact (26) and the second tracking-device contact (28) when the tracking device (12) is operatively received in the device-receiving volume (16).

22. The housing (10) of claim 1, further comprising the tracking device (12) operatively received in the device-receiving volume (16).
